Sparse-Dense Side-Tuner for efficient Video Temporal Grounding

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Abstract

Video Temporal Grounding (VTG) involves Moment Retrieval (MR) and Highlight Detection (HD) based on textual queries. For this, most methods rely solely on final-layer features of frozen large pre-trained backbones, limiting their adaptability to new domains. While full fine-tuning is often impractical, parameter-efficient fine-tuning –and particularly side-tuning (ST)– has emerged as an effective alternative. However, prior ST approaches this problem from a frame-level refinement perspective, overlooking the inherent sparse nature of MR. To address this, we propose the Sparse-Dense Side-Tuner (SDST), the first anchor-free ST architecture for VTG. We also introduce the Reference-based Deformable Self-Attention, a novel mechanism that enhances the context modeling of the deformable attention –a key limitation of existing anchor-free methods. Additionally, we present the first effective integration of InternVideo2 backbone into an ST framework, showing its profound implications in performance. Overall, our method significantly improves existing ST methods, achieving highly competitive or SOTA results on QVHighlights, TACoS, and Charades-STA, while reducing up to a 73% the parameter count w.r.t. the existing SOTA methods. The code is publicly accessible at https://github.com/davidpujol/SDST.

1. Introduction

Recently the field of Video Understanding has gained attention due to its potential in applications like search engines or recommendation systems. A key task in this field is Video Temporal Grounding (VTG), which localizes moments within videos based on textual descriptions. Typically, this involves doing both Moment Retrieval (MR) [2, 5, 7, 11, 15] and Highlight Detection (HD) [1, 8, 10, 33, 40]. More specifically, MR predicts moment boundaries w.r.t. textual queries, while HD offers a more interpretable perspective, predicting frame-level saliency scores.

In the past, existing works focused on either MR or HD, given the lack of datasets that combined both tasks. QVHighlights [15] shifted this paradigm, proposing the first

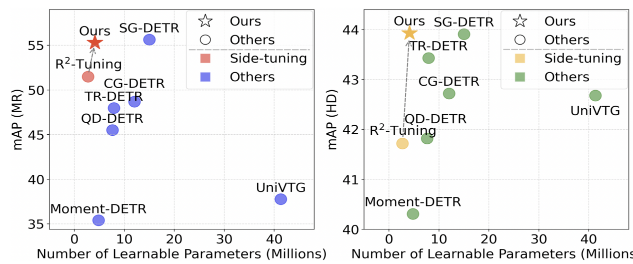


Figure 1. Comparison of our proposed method and the main MR (left) and HD (right) baselines. These are all evaluated on QVHighlights *val* split and use InternVideo2-1B features. These results show that our method improves existing side-tuning works and attains SOTA results while significantly reducing the number of trainable parameters.

dataset suitable for this multi-task setup. Since then, various works have been proposed following *anchor-based* [2, 17, 21, 22], *anchor-free* [7, 15, 25] and even LLM-based [23, 24] approaches. All of these, however, incur in a common critical limitation, which is relying on the final-layer features of frozen backbones. This is especially limiting when facing large distribution shifts between the pre-training distribution and that of the downstream task [29]. This issue is particularly pronounced in VTG, where an image-domain backbone [31] is transferred to the video domain.

A typical solution is fine-tuning; however, full fine-tuning is impractical due to its high computational cost. Doing this more efficiently has thus become critical, motivating the interest in *parameter-efficient fine-tuning (PEFT)* [9] methods –e.g., Prompting or Adapters– which optimize only a small subset of parameters. Unfortunately, these still require full back-propagation through the backbone, making it still memory-intensive. To address this, [34] introduced *side-tuning* (ST), an effective PEFT, and *memory-efficient fine-tuning* (MEFT) method. ST creates a parallel pathway to refine intermediate features, thus back-propagating over a minimal set of parameters only. In this regard, we highlight the work of R^2-Tuning [21], the first ST approach for VTG, which recursively fuses multi-modal intermediate CLIP [31] embeddings. [21] also proposes adapting these frame-level embeddings for MR by generating a dense set of anchors. As shown in Fig. 1, this method proves ineffective for MR – a highly sparse task where videos may contain very few ground-truth action.

This motivates our dual-stream *Sparse-Dense Side-Tuner (SDST)*, the first proposal-free ST method, carefully designed for multi-task learning across sparse (MR) and dense (HD) tasks. For this, SDST jointly refines frame-level embeddings –suitable for HD– while learning what we call the *recurrent decoder queries* for MR. We also identify an implicit contextual limitation of the deformable attention [45] module of anchor-free architectures like ours, which in our task results in collapsing offsets around their initialization values. This also impedes the selection of keys outside of the current estimated moment boundaries, critical to potentially refine the boundaries of longer moments. Consequently, we propose the alternative *Reference-based Deformable Self-Attention (RDSA)*, which naturally addresses this issue by reformulating the CA into a SA-based mechanism. Finally, we tackle the performance degradation of ST methods derived from the use of image-based CLIP, over more advanced spatio-temporal VLMs [36], as noted in [7]. A key challenge of exploiting [36] is, however, defining an effective token pooling strategy, as naive strategies like CLS pooling yield significant performance drops. This motivates our proposed module re-utilization scheme, which results in the first successful integration of InternVideo2 [36] into an ST framework.

In summary, our main contributions are threefold: 1) We propose SDST, the first anchor-free ST architecture, specifically tailored for complex sparse-dense multi-task setups like VTG. Our method significantly outperforms existing ST architectures on QVHighlights [15], TACoS [32] and Charades-STA [5] (see Fig. 1). SDST also performs competitively and even surpasses existing SOTA while reducing its learnable parameter count by up to 73%, and incurring in a minimal memory overhead. 2) We identify a key contextual limitation of the deformable attention mechanism that, in our task, results in collapsing offsets. Consequently, we propose RDSA, which tackles this issue by allowing more complex key-selection strategies. 3) We address the limitations of transfer learning from an image-domain backbone to a video domain by integrating, for the first time, the more advanced spatio-temporal backbone [36] into an ST framework. This proves to be a non-trivial challenge with massive performance implications.

2. Related work

Video temporal grounding (VTG): One relevant task in Video Understanding is VTG [17] which aims to identify actions from text queries. Traditionally, this task was approached either from a MR [2, 5, 7, 11, 15, 25, 26] or HD [1, 8, 10, 33, 40] perspective. MR focuses on predicting specific action proposals while HD aims to compute the saliency scores for each of the frames w.r.t. the query. Nevertheless, after the proposal of QVHighlights [15], the literature started approaching this from a multi-task learning perspective. In this regard, arguably the most notable family of methods is that based on DETR [7, 14, 15, 25, 26, 39] given the sparse nature of MR. These methods normally fuse both video and textual modality into a final embedding which is used for HD and as a memory of a Transformer decoder that refines a set of learnable queries. More in detail, Moment-DETR [15] constitutes the first baseline, which uses a standard Transformer encoder to process both modalities simultaneously. QD-DETR [26] proposes instead to inject the textual information to the video modality with cross attention (CA), followed by a temporal modeling module. CG-DETR [25] reduces the negative impact of irrelevant textual tokens by adding an Adaptive CA module that leverages dummy tokens to redirect attention weight. In this same line, SG-DETR [7] introduces a saliency-guided CA, weighting the contribution of irrelevant textual tokens based on the computed saliency scores. Despite the notable success of these proposal-free methods, one core limitation that they all share is that they rely solely on the last-layer features of a frozen CLIP [31] and/or Slowfast [4] backbones. [41] overcomes this limitation via the full fine-tuning of CLIP, even though this becomes nearly intractable given the resources that this requires. R^2-Tuning [21], the most similar method to our work, overcomes this limitation by proposing the first PEFT and MEFT method for VTG, which recursively applies a multi-modal fusion ST. In this work, we argue that one of its core drawbacks, however, is its anchor-based nature, remaining oblivious to the shown benefits of DETR-based architectures in very sparse tasks like MR [7, 16, 20]. This motivates our proposed SDST, which incorporates the best of both PEFT and MEFT.

Parameter-and-memory-efficient fine-tuning: Foundation models and VLMs [31, 37] have become the cornerstone of recent advances in Video Understanding applications. The use of pre-extracted features remains, however, an important limitation to its application on downstream tasks like VTG [17]. Fine-tuning these huge models is often simply infeasible given the large resources that these require. This has motivated the rise of PEFT [9], which aims to democratize the fine-tuning process by restricting the tunable parameters to the bare minimum. In this regard, some of the most prominent approaches are Prompting [12, 44] and the use of Adapters [6, 28]. The first keeps the backbone frozen while learning prompts that are appended to the input, bridging the gap between the backbone's expected distribution and that of the downstream task. Adapters, in contrast, incorporate small trainable modules in the backbone while keeping the rest unchanged. These methods, however, still require full backpropagation through the frozen model, making them memory-inefficient. Recently, the new paradigm of *ST* [19, 30, 34] –and particularly [21] for VTG– has gained attraction creating a parallel pathway to exploit intermediate backbone representations while guaranteeing that backpropagation is only applied to this small parallel module. Interestingly, these methods normally rely on pre-extracted CLIP representations for tasks like HD/MR. As shown by [7], this is a very limiting aspect given the lack of temporal reasoning of CLIP or its difficulties in understanding textual queries beyond simple spatial descriptions. For this reason, in this work, we propose a novel ST architecture that, for the first time, relies on the more advanced InternVideo2 [36] backbone. Unlike CLIP, this backbone is trained on video-domain inputs, resulting in enhanced spatial and temporal modeling capabilities.

Deformable attention: A key component of most SOTA methods for VTG like SG-DETR [7] is their deformable attention module [38, 45]. This module addresses the well-studied slow convergence of DETR [3]. It does so by limiting the selectable keys based on the predictions of the learnable queries, completely decoupling both query and key spaces. Despite desirable in terms of efficiency, it has been previously noted in the literature [43] that the lack of context of the queries w.r.t. the key/value space can lead to suboptimal attendable key selection. VTG methods often mitigate this effect by partially or completely initializing the learnable queries based on the DETR-memory [43] –i.e., frame-level representations in our case. In this work, we empirically demonstrate the limitations of these initialization-based methods for VTG tasks, which motivates our proposed simple yet effective alternative, the *Reference-based Deformable Self-Attention (RDSA)*. This reformulates the deformable CA into a deformable SA, naturally solving the aforementioned issue.

3. Method

3.1. Problem definition

In this paper, we address the problem of MR and HD based on textual descriptions. For this, we consider an arbitrary input-query pair $(\mathbf{X}^v, \mathbf{X}^t)$ where $\mathbf{X}^v \in \mathbb{R}^{T\times H\times W\times 3}$ and $\mathbf{X}^t \in \mathbb{R}^{L\times F_e}$. Here T and L are the number of frames and tokens, and F_e the textual encoding dimension. Our goal is to predict, for the given video-query pair, its respective frame-wise saliency scores $\mathbf{Y}^s \in \mathbb{R}^T$ for HD, and the M action moments $\mathbf{Y}^m \in \mathbb{R}^{M\times 2}$ for MR.

3.2. Overview

In our work we propose a dual-stream ST architecture that includes a dense (frame-level) and a sparse (segment-level) stream for MR and HD respectively (see Fig. 2a). More specifically, we first leverage a frozen InternVideo2-1B [36] backbone, a model that possesses powerful spatio-temporal modeling capabilities, to extract K intermediate visual-textual representations. These are then processed by multiple weight-sharing SDST layers, which refine dense representations and recurrent decoder queries. More in detail, at each level, the model refines the dense embeddings via multi-modal textual conditioning as well as temporal modeling. This signal is then used to condition the sparse stream $\mathcal{S}$, that builds on [16] while incorporating our novel deformable attention mechanism, namely *Reference-based Deformable Self-Attention* (see Fig. 2b). This mitigates the shortcomings of the deformable CA mechanism, enhancing the contextual information of the decoder queries and thus improving the quality of the selected keys. After unrolling this process K times, we apply the dense and the sparse prediction heads to compute the saliency scores and the predicted segment boundaries, respectively.

3.3. Sparse-Dense Side-Tuner (SDST)

3.3.1. Recurrent refinement with side-tuners

Our work builds on an ST framework, which first requires extracting K intermediate video and textual features $\tilde{\mathbf{V}} \in \mathbb{R}^{K\times T\times F_v}$ and $\tilde{\mathbf{T}} \in \mathbb{R}^{K\times L\times F_t}$ (find the details in Sec. 4). Here F_v and F_t are their respective dimensionalities. We then define a zero-initialized dense embedding $\mathbf{D}^0 \in \mathbb{R}^{T\times F}$ and a set of M learnable recurrent moment proposals inspired by [20], which we refer to as *recurrent decoder queries*. The queries include the learnable center-width moment references $\mathbf{R}^0 \in \mathbb{R}^{M\times 2}$, and their corresponding latent embeddings $\mathbf{H}^0 \in \mathbb{R}^{M\times F}$. We then define the recurrence for a given level $1 \leq \ell \leq K$ as follows:

$$\mathbf{D}^{\ell+1}, \mathbf{R}^{\ell+1}, \mathbf{H}^{\ell+1} = SDST(\mathbf{D}^{\ell}, \mathbf{R}^{\ell}, \mathbf{H}^{\ell}, \tilde{\mathbf{V}}^{\ell}, \tilde{\mathbf{T}}^{\ell}), \quad (1)$$

where SDST represents the shared-weight layers of our model, progressively refining the representations across two streams: the *1) dense learning stream* $\mathcal{D}$ and the *2) sparse learning stream* $\mathcal{S}$, that we describe below.

3.3.2. Dense learning stream $\mathcal{D}$

Following [21], this stream first refines the dense embeddings $\mathbf{D}^\ell$ by conditioning it to the video-textual embeddings $\tilde{\mathbf{V}}^\ell$ and $\tilde{\mathbf{T}}^\ell$. For this, both of these multi-modal embeddings are projected to a shared F-dimensional space, using two MLPs ($\mathcal{F}_v : F_v \to F$ and $\mathcal{F}_t : F_t \to F$):

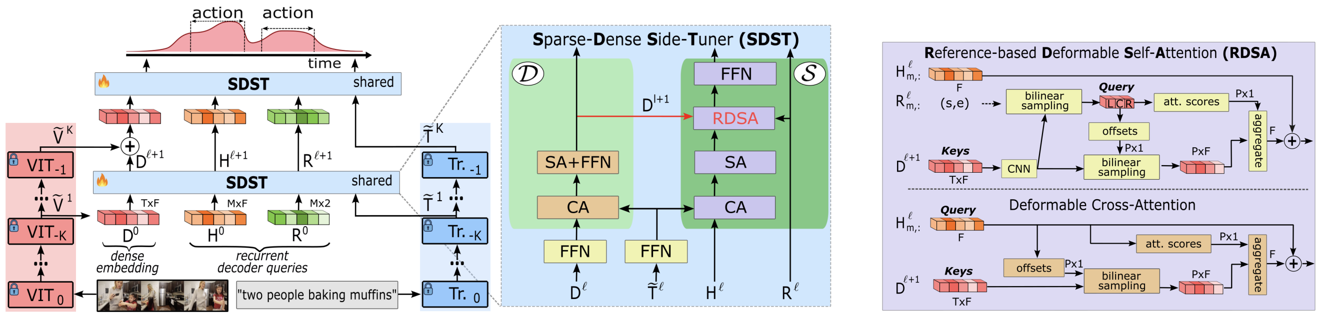
$$\mathbf{V}^\ell = \mathcal{F}_v(\tilde{\mathbf{V}}^\ell)\ ,\ \mathbf{T}^\ell = \mathcal{F}_t(\tilde{\mathbf{T}}^\ell). \quad (2)$$

Then we incorporate the clip-wise visual information to $\mathbf{D}^\ell$ via a weighted sum modulated by $\beta^\ell \in [0, 1]$, a zero-initialized layer-dependent parameter that yields:

$$\mathbf{D}^\ell := \beta^\ell \mathbf{D}^\ell + (1 - \beta^\ell)\mathbf{V}^\ell. \quad (3)$$

Thereafter, we inject the textual information with CA, where $\mathbf{T}^\ell$ does not include the CLS token. We also apply a temporal modeling module defined as a SA and a subsequent point-wise feed forward network (PFFN):

$$\mathbf{D}^{\ell+1} = PFFN(SA(CA(\mathbf{D}^\ell, \mathbf{T}^\ell, \mathbf{T}^\ell))), \quad (4)$$



(a) Overview of our proposed architecture.

(b) RDSA vs Def.CA[38] for a proposal m.

Figure 2. Our method (left) first processes the video and textual inputs using [36], and then recursively applies a shared SDST, a recurrent dual-stream model attached to the last K layers. This iteratively refines dense embeddings and learnable *recurrent decoder queries*. The SDST consists of a dense stream $\mathcal{D}$ for temporal and multi-modal dense refinement, and a sparse stream $\mathcal{S}$ that refines the recurrent queries conditioned on the dense signal. $\mathcal{S}$ includes the RDSA module (right), a context enhanced deformable attention mechanism that restricts the selectable keys based on the center, left-, and right-most action embeddings. The final dense embedding solves HD, while the recurrent queries address MR.

3.3.3. Sparse learning stream $\mathcal{S}$

This stream can be seen as a recurrent DETR-based mechanism that exploits the complementarity of both sparse (MR) and dense (HD) tasks to refine the recurrent decoder queries. This is, refining the center-width references $\mathbf{R}^\ell \in \mathbb{R}^{M\times 2}$ and their latent embeddings $\mathbf{H}^\ell \in \mathbb{R}^{M\times F}$. Similar to the stream $\mathcal{D}$, we first incorporate a CA module that conditions $\mathbf{H}^\ell$ to the textual query. This is followed by a SA module to enable the information flow between different moment proposals:

$$\mathbf{H}^\ell = SA(CA(\mathbf{H}^\ell, \mathbf{T}^\ell, \mathbf{T}^\ell)). \tag{5}$$

Another essential aspect of this stream is effectively conditioning the recurrent decoder queries to the video modality. Existing works [7, 16, 20] typically rely on the use of deformable attention due to its convergence and performance benefits [38, 45]. This mechanism leverages attention queries to predict offsets that define a limited set of selectable keys. However, in this task we observe that 1) these offsets collapse near their initialization and 2) are unable to *look* beyond the current estimated boundaries (see Sec. 6.2). We attribute this to contextual limitations, which motivates our proposed alternative *Reference-based Deformable Self-Attention* (RDSA), enabling an enhanced cross-modality injection (see Sec. 3.3.4). We also incorporate a simple PFFN to enhance the model's expressivity:

$$\mathbf{H}^\ell = PFFN(RDSA(\mathbf{R}^\ell, \mathbf{H}^\ell, \mathbf{D}^{\ell+1})). \tag{6}$$

Importantly, all of these modules are shared across the K levels for parameter efficiency, and include residual connections and DropPath [13] for stability and regularization.

3.3.4. Reference-based deformable self-attention

Deformable attention in the context of existing anchor-free methods: One of the keys to the success of recent anchor-free methods like DETR –including works tackling VTG [7]– is the use of deformable attention mechanisms [45] for cross-modal interaction. This mechanism is known to provide faster convergence speed, reduced time complexity, and as shown in Sec. K, overall better performance than standard CA. As depicted in Fig. 2b (down), the key of this module is the lack of explicit interaction between queries and keys, which are defined as

$$\mathbf{Q} = \mathbf{X}_{\mathcal{Q}} \mathbf{W}^{def}_{\mathcal{Q}}\ ,\ \mathbf{K} = \mathbf{X}_{\mathcal{K}} \mathbf{W}^{def}_{\mathcal{K}}, \tag{7}$$

where $\mathbf{W}^{def}_{\mathcal{Q}}$, $\mathbf{W}^{def}_{\mathcal{K}}$ are two linear projections. The deformable attention thus avoids computing the $\mathbf{QK}^T$ similarity matrix, and instead employs an offset and attention-score predictors, $\mathcal{G}_\Delta$ and $\mathcal{G}_\mathbf{A}$ to select and weight –only based on the queries– a small subset P of selectable keys. The output of this mechanism, the weighted aggregation of the selected keys, namely $\mathbf{S}$, is computed as follows

$$\Delta = \mathcal{G}_\Delta(\mathbf{Q}) \in \mathbb{R}^{M\times P},\ \mathbf{A} = \mathcal{G}_\mathbf{A}(\mathbf{Q}) \in \mathbb{R}^{M\times P}, \tag{8}$$

$$\mathbf{S} = \sum_{p=1}^{P} (\mathbf{A}_{:,p}\ \mathbf{K}[\mathbf{c} + \mathbf{w} \odot \Delta_{:,p}]) \in \mathbb{R}^{M\times F} \tag{9}$$

where $\mathcal{G}_\Delta$, $\mathcal{G}_\mathbf{A}$ are often modeled as CNNs and $x[y]$ is the bilinear sampling of x on index(es) y. Notably, in our 2-dimensional setup, the offsets $\Delta_{:,p}$ are added to center reference $\mathbf{c} = \mathbf{R}^\ell_{:,0} \in \mathbb{R}^M$, and weighted over the action width $\mathbf{w} = \mathbf{R}^\ell_{:,1} \in \mathbb{R}^M$. See Sec. K for additional details.

Limitations: Interestingly, in the literature we find that when it comes to the deformable attention, most works operate with a certain independence to whether they deal with a deformable self-attention or cross-attention, assuming that the benefits of this mechanism extrapolate to both scenarios. In this work, however, we find that in fact, the deformable attention is only naturally suited for <u>self-attention</u> scenarios as its effectiveness inherently relies on *implicit interactions* between $\mathbf{Q}$ and $\mathbf{K}$ –not present in the cross-attention case. Concretely, in a self-attention setup, $\mathbf{X}_\mathcal{Q} = \mathbf{X}_\mathcal{K}$ (see Eq. 7). Thus, modeling $\mathcal{G}_\Delta$ and $\mathcal{G}_\mathbf{A}$ as CNNs naturally allows queries to gain context of the local neighborhood of queries, and consequently, of the key/value space. This breaks their independence assumption, giving the queries critical information to decide *where to look*.

Importantly, this is not the case with the deformable cross attention where we try to inject cross-modal information from the keys to the queries, which is our goal in this this work. In this scenario, $\mathbf{X}_\mathcal{Q} \neq \mathbf{X}_\mathcal{K}$, and thus, using CNN-based offset predictors does not inject knowledge of the key/value space, making its guesses *blind* or based on the *blind* predictions of previous iterations. Concretely, in Sec. 6.2 we show that in our task, the drawbacks of this naive deformable cross-attention have various empirical implications such as the offset collapse near the offset initialization values, or the inability to select keys from frames beyond the estimated action boundaries, critical to refine segments into longer actions. This motivates our proposed *Reference-based Deformable Self-Attention*, an alternative mechanism that allows reformulating the deformable CA module into a deformable SA based on the references $\mathbf{R}^\ell$.

Our proposed alternative (RDSA): Our proposed RDSA (see Fig. 2b) takes as input the center-width references $\mathbf{R}^\ell \in \mathbb{R}^{M\times 2}$ and the dense embeddings $\mathbf{D}^\ell \in \mathbb{R}^{T\times F}$. Then, differently from other methods that leverage the learnable queries $\mathbf{Q} = \mathbf{H}^\ell$ to compute the offsets and attention scores (see Eq. 8), we propose using an alternative query embedding. Concretely, we first refine the dense embedding $\mathbf{D}_l$ with a simple CNN, allowing frames to gain local context of their surroundings. Then, we use bilinear sampling to extract three key action embeddings: left-most (l), center (c), and right-most (r). The center provides highly informative action features, while the extremes help refine the moment boundaries. This yields,

$$\hat{\mathbf{Q}} = \hat{\mathbf{X}}_\mathcal{Q}\mathbf{W}_\mathcal{Q}^{def}\ ,\ \hat{\mathbf{X}}_\mathcal{Q} = CNN(\mathbf{D}^\ell)[l,c,r]. \tag{10}$$

where $\hat{\mathbf{X}}_\mathcal{Q} \in \mathbb{R}^{M\times 3F}$. This rewrites Eq. 8 to

$$\hat{\Delta} = \mathcal{G}_\Delta(\hat{\mathbf{Q}}),\ \hat{\mathbf{A}} = \mathcal{G}_\mathbf{A}(\hat{\mathbf{Q}}). \tag{11}$$

Finally, we apply the deformable attention from Eq. 9 using the new offsets and attention scores $\hat{\Delta}$ and $\hat{\mathbf{A}}$. This can be viewed as *asking the key spots of the action where to look* based on the textual information that we previously injected in Eq. 5. This naturally solves the context limitation of the standard deformable CA given that $\hat{\mathbf{X}}_\mathcal{Q}$ and $\mathbf{X}_\mathcal{K}$ now live in the same latent space –as they are both derived from $\mathbf{D}^\ell$. This allows the model to make informed decisions about which keys to attend to while keeping all its core computational advantages.

3.4. Prediction heads and training objectives

After recursively applying the SDST over K intermediate layers, we compute the respective saliency scores and action segment proposals, as well as define their corresponding objective functions. We briefly describe these below, but refer interested readers to Sec. C for more details.

Highlight Detection: To solve HD, we define the saliency scores as the cosine similarity between the dense embedding $\mathbf{D}^K$ and a pooled representation of $\mathbf{T}^K$. We then apply an InfoNCE loss [27] that learns a ranking of these scores

$$\mathcal{L}_{HD} = \lambda_0\mathcal{L}_{InfoNCE}. \tag{12}$$

Moment Retrieval: To solve MR we first apply the Hungarian algorithm to obtain a one-to-one matching between the predicted queries and the GT actions. We then project $\mathbf{H}^\ell$ to compute the action probability $\mathbf{p} \in \mathbb{R}^M$ of the M different queries, which we train with a FocalLoss [18]. We also compute the moment boundaries $\hat{\mathbf{Y}}^m \in \mathbb{R}^{M\times 2}$ which we optimize using an L1 and an IoU loss. Finally, we use an L1 loss to learn an actionness score, which predicts the maximum IoU of each query w.r.t. the GT actions

$$\mathcal{L}_{MR} = \lambda_1\mathcal{L}_{act} + \sum_{l=1}^{\ell}\lambda_2\mathcal{L}_{cls}^\ell + \lambda_3\mathcal{L}_{l1}^\ell + \lambda_4\mathcal{L}_{IoU}^\ell. \tag{13}$$

Note that all but the $\mathcal{L}_{act}$ term are optimized across different refinement levels to promote a faster convergence.

Alignment losses: To improve the video-textual alignment, we define $\mathcal{L}_{align}$ based on two SampledNCE losses [42] to align $\mathbf{V}$ and $\mathbf{T}$ embeddings across the different refinement levels. We apply this along both the batch and the intermediate-layer dimensions. This pulls the action embeddings closer to the textual representations while learning complementary information across intermediate layers.

Final loss: The final loss is defined as

$$\mathcal{L} = \lambda_5\mathcal{L}_{HD} + \lambda_6\mathcal{L}_{MR} + \lambda_7\mathcal{L}_{align}. \tag{14}$$

4. Extracting intermediate representations of InternVideo2

The cornerstone of ST is leveraging intermediate multimodal representations from frozen VLMs. While previous ST approaches predominantly rely on CLIP, we argue that CLIP has notable limitations in temporal modeling and aligning visual features with textual queries beyond simple static descriptions. To overcome these limitations, we incorporate InternVideo2 [36] into our framework, leveraging its advanced spatio-temporal modeling capabilities [7]. However, extracting intermediate representations from InternVideo2 –particularly from its visual encoder $\mathcal{E}_v$– remains a challenging and underexplored task in the literature, despite its significant impact on performance.

In particular, the main challenge arises from how to pool the spatio-temporal token dimension L_v of $\hat{\mathbf{V}}^\ell \in \mathbb{R}^{T\times L_v\times F}$, the output embedding of the ℓ-th layer of $\mathcal{E}_v$, into the final intermediate features $\tilde{\mathbf{V}}^\ell \in \mathbb{R}^{T\times F}$ (see Eq. 1). Prior CLIP-based methods simply use the CLS token as "summaries" over L_v. However, we find this strategy to be suboptimal for InternVideo2 (see Sec. 6.1) as this limits its spatial-aggregation capabilities. Notably, [36] optimizes an AdaptivePool module that computes the final-layer embeddings, which are then aligned with the text. We conjecture that leveraging this enhanced multi-modal alignment is key to performance. Unfortunately, such pooled representation is only computed at the last layer, leaving the question of *how to pool the remaining intermediate layers.* Ideally, we would optimize layer-independent AdaptivePool modules, but this would require full back-propagation through the entire model. A more efficient alternative would be computing gradients of the AdaptivePool modules only. This is, however, still computationally infeasible, not for the gradients themselves, but because of the need to load L_v spatio-temporal tokens per frame. This increase in the input memory size translates, for instance, to a $15\times$ memory increase for QVHighlights. We thus hypothesize that re-using this frozen pooling module across the K intermediate layers allows a memory efficient training –no additional backpropagation nor memory requirements are needed– while leveraging its enhanced pooling capabilities, even despite their respective distribution shifts. Formally, we compute

$$\tilde{\mathbf{V}}^{\ell} = AdaptivePool\left(\hat{\mathbf{V}}^{\ell}\right) \in \mathbb{R}^{T\times F},\ 1 \leq \ell \leq K. \quad (15)$$

As shown in Sec. 6.1, this pooling strategy considerably improves other alternatives, demonstrating the benefits of this module re-utilization.

5. Experimentation

5.1. Experimental setup

We test our proposal on various datasets for MR and HD, namely QVHighlights [15], TACoS [32], and the Charades-STA [5] dataset. Further details of the datasets can be found in Sec. B. For the task of MR on QVHighlights, we compute Recall@1 with two IoU thresholds, 0.5 and 0.7, and the mean average precision (mAP) at IoU thresholds between 0.5 and 0.95 with a step of 0.05 –i.e., [0.5:0.05:0.95]. For HD, we report mAP and HIT@1 over the positive frames –i.e., the most salient ones classified as ”VeryGood”. For TACoS and Charades-STA, we compute Recall@1 at 0.3, 0.5, and 0.7 IoU thresholds, as well as mIoU.

5.2. Main experimental results

We begin by comparing the performance of SDST over various relevant baselines. Concretely, observe in Tab. 1 the evaluation of our method on the *test* and *val* splits of QVHighlights. These results indicate that our method considerably outperforms R^2-Tuning. Similarly, SDST performs very competitively and even surpasses on several metrics the current SOTA method, namely SG-DETR [7]. Note however that our method attains this performance while having only a 27% of the parameter count of SG-DETR. We refer interested readers to Sec. M for the statistical significance analysis.

Most of the presented methods in Tab. 1 rely on alternative backbones different from InternVideo2 –e.g., CLIP or Slowfast. To establish a more fair comparison, in Fig. 1 we follow the work of [7] and evaluate a set of relevant baselines when using InternVideo2 features –find the complete ablation in Sec. E. Observe that our work performs on par with the SG-DETR and even more importantly to our work, the SDST very considerably outperforms R^2-Tuning, improving the MR performance by a 3.82% average mAP or the HD by a 2.21% mAP. This empirically demonstrates the benefits of leveraging a proposal-free architecture over a more rigid anchor-based method. Also find in Sec. H an extended comparison with other ST works.

To complement our findings, in Tab. 2 we evaluate our method on the Charades-STA and TACoS benchmarks. Observe that our method obtains SOTA in both of these datasets. In Charades-STA, for instance, improving by 2.71% of R1@0.7 and 2.06% of mIoU, the existing SOTA. Similarly, in TACoS, SDST improves SG-DETR by 2.39% R1@0.7 and 1.27% mIoU.

6. Ablation studies

In this section, we ablate over various relevant aspects of our main contributions. Note that unless states otherwise, we follow the literature and evaluate them on the *val* split of QVHighlights. Also, we write write in **bold** the best results.

6.1. Leveraging InternVideo2 features for ST

1) Does pooling matter? As argued in Sec. 4, one of the main challenges in using the InternVideo2 backbone for ST is pooling the spatio-temporal clip-wise intermediate representations. In Tab. 3 we compare different pooling strategies. Concretely, we compare the standard CLS-pooling, average-pooling, and our proposed re-utilization of the frozen AdaptivePool (Sec. 4). Observe that CLS-pooling results in a considerable performance degradation of up to 5.07% average mAP on MR or 7.74% HIT@1 on HD w.r.t. the adaptive pooling strategy. The average pooling partially mitigates this degradation, but still obtains a considerably decreased performance. This confirms the significant impact of a carefully chosen pooling strategy.

2) Feature refinement vs. feature sampling: One of the most critical decisions of ST is determining the number of intermediate levels that should be used. Normally, doing so involves evaluating the performance when refining the last K intermediate layers [21] . As shown in Fig. 3 (red) this indicates that in our case, the best performing K is 4. The literature often suggests this argument to be compelling enough to claim that using intermediate representations is beneficial over using only last-layer features. In this work, however, we find it reasonable to wonder: *Does this analysis depict the importance of intermediate features, or is it simply evidencing the need to do multiple refinement steps,*

Method	test							val							#Params
	MR					HD		MR					HD		
	R1		mAP			≥ Very good		R1		mAP			≥ Very good		
	@0.5	@0.7	@0.5	@0.75	Avg.	mAP	HIT@1	@0.5	@0.7	@0.5	@0.75	Avg.	mAP	HIT@1	
BeautyThumb	–	–	–	–	–	14.36	20.88	–	–	–	–	–	–	–	–
DVSE	–	–	–	–	–	18.75	21.79	–	–	–	–	–	–	–	–
MCN	11.41	2.71	24.94	8.22	10.67	–	–	–	–	–	–	–	–	–	–
CAL	25.49	11.54	23.40	7.65	9.89	–	–	–	–	–	–	–	–	–	–
XML+	46.69	33.46	47.89	34.67	34.90	35.38	55.06	–	–	–	–	–	–	–	–
Moment-DETR	52.89	33.02	54.82	29.40	30.73	35.69	55.60	53.94	34.84	–	–	32.20	35.65	35.65	4.8M
UMT	56.23	41.18	53.83	37.01	36.12	38.18	59.99	60.26	44.26	56.70	39.90	38.59	39.90	64.20	14.9M
MomentDiff	58.21	41.48	54.57	37.21	36.84	–	–	–	–	–	–	–	–	–	–
QD-DETR	62.40	44.98	62.52	39.88	39.86	38.94	62.40	62.68	46.66	62.23	41.82	41.22	39.13	63.03	7.6 M
MH-DETR	60.05	42.48	60.75	38.13	38.38	38.22	60.51	60.84	44.90	60.76	39.64	39.26	38.77	61.74	8.2 M
UniVTG	58.86	40.86	57.60	35.59	35.47	38.20	60.96	59.74	–	–	–	36.13	38.80	61.8	41.3M
TR-DETR	64.66	48.96	63.98	43.73	42.62	39.91	63.42	67.10	51.48	66.27	46.42	45.09	–	–	7.9 M
CG-DETR	65.43	48.38	64.51	42.77	42.86	40.33	66.21	67.35	52.06	65.57	45.73	44.93	40.80	66.70	12.0 M
BAM-DETR	62.71	48.64	64.57	46.33	45.36	–	–	65.10	51.61	65.41	48.56	47.61	–	–	–
EaTR	–	–	–	–	–	–	–	61.36	45.79	61.86	41.91	41.74	37.15	58.65	9.0 M
Mr. BLIP	74.77	60.51	68.12	53.38	–	–	–	76.13	63.35	69.39	55.78	–	–	–	19.0 M
LLaVA-MR	76.59	61.48	69.41	54.40	–	–	–	78.13	64.13	69.64	56.32	–	–	–	17.0 M
HL-CLIP	–	–	–	–	–	41.94	70.60	–	–	–	–	–	42.37	72.40	2.0 M
R^2-Tuning	68.03	49.35	69.04	47.56	46.17	40.75	64.20	68.71	52.06	–	–	47.59	40.59	64.32	2.7 M
SG-DETR†	**72.20**	**56.60**	**73.20**	**55.80**	**54.10**	**43.76**	**69.13**	–	–	**73.52**	**57.91**	**55.64**	<u>43.91</u>	<u>71.47</u>	15.0 M
Flash-VTG†	70.69	53.96	<u>72.33</u>	53.85	52.00	–	–	73.10	57.29	72.75	54.33	52.84	–	–	10.9 M
Ours†	<u>70.82</u>	<u>56.23</u>	71.31	<u>54.99</u>	<u>53.31</u>	43.40	**69.13**	**73.68**	**60.90**	**73.52**	<u>57.42</u>	<u>55.60</u>	**44.00**	**72.00**	4.1 M

Table 1. MR and HD results for QVHighlights *test* and *val* split.† indicates the use of InternVideo2 features. **Bold** stands for the best and <u>underline</u> for the second-best, considering only works supporting both MR and HD. The rest are marked in gray.

Method	Charades-ST			TACoS		
	R@0.5	R@0.7	mIoU	R@0.5	R@0.7	mIoU
M-DETR	53.6	31.4	-	24.7	12.0	25.5
UMT	48.3	29.3	-	-	-	-
UniVTG	58.0	35.7	50.1	35.0	17.4	33.6
QD-DETR	57.3	32.6	-	36.8	21.1	35.8
CG-DETR	58.4	36.3	50.1	39.6	22.2	36.5
BAM-DETR	60.0	39.4	52.3	41.5	26.8	39.3
TR-DETR	57.6	33.5	-	-	-	-
MR. BLIP	69.3	49.3	58.6	-	-	-
LLaVA-MR	<u>70.6</u>	49.6	<u>59.8</u>	-	-	-
R^2-Tuning	59.8	37.0	50.9	38.7	25.1	35.9
SG-DETR†	70.2	49.5	59.1	**44.7**	<u>29.9</u>	<u>40.9</u>
FlashVTG†	70.3	<u>49.9</u>	–	41.8	24.7	37.6
Ours†	**72.0**	**52.6**	**61.2**	<u>44.5</u>	**32.3**	**42.2**

Table 2. Comparison on Charades-STA and TACoS datasets. † indicates the use of InternVideo2 features. **Bold** stands for the best and <u>underline</u> for the second-best.

Pool strat.	MR					HD	
	R1@0.5	R1@0.7	mAP@0.5	mAP@0.75	mAP	mAP	HIT@1
CLS	66.45	52.65	67.96	51.32	50.53	41.01	64.26
Avg. pool	70.65	56.9	70.43	54.58	53.44	43.06	69.68
Adapt. pool	**73.68**	**60.90**	**73.52**	**57.42**	**55.60**	**44.00**	**72.00**

Table 3. Effect of using different pooling strategies.

regardless of the chosen features?

To shed light on this issue, in Fig. 3 (blue) we evaluate the performance when doing K different refinement steps, but importantly, we always use the <u>last-layer features</u> only. Observe that for $K = 2$ and $K = 3$, actually, this improves the homologous experiments with intermediate features. Differently, for $K = 4$ and $K = 5$, the use of last-layer features harms the performance. These experiments thus suggest that proving the usefulness of intermediate features is not as straightforward as previously thought. In fact, these indicate that the advantages of using intermediate features arise only as we consider shallower layers.

3) Why not sampling features from even shallower layers? As shown in Tab. 4, this is not necessarily helpful

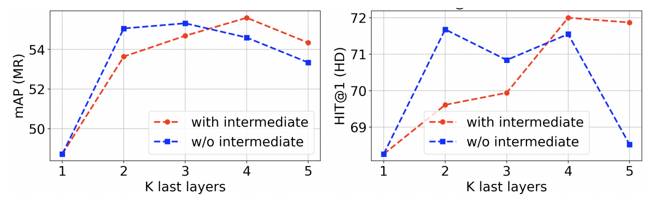


Figure 3. Ablation of the number of refinement levels and of their use of intermediate or last-layer features only.

Levels	MR					HD	
	R1@0.5	R1@0.7	mAP@0.5	mAP@0.75	mAP	mAP	HIT@1
[37,38,39,40]	73.68	60.90	73.52	57.42	55.60	44.0	72.00
[25,30,35,40]	71.48	59.48	71.45	56.37	54.83	43.42	70.19
[10,20,30,40]	69.03	55.35	69.56	53.37	52.47	42.73	68.39

Table 4. Ablation of different sampling strategies of intermediate features. The first column indicates the $K = 4$ sampled layers –of a total of 40 layers.

because of what we call, *depth-pooling trade-off*. We conjecture that sampling from shallower layers does in fact provide additional complementary information. Nevertheless, this inevitably results in a distribution shift w.r.t. the last-layer feature distribution, hindering the effectiveness of the frozen AdaptivePool (see Eq. 15). The infeasibility of retraining this module for computational reasons, thus, leaves us with a delicate trade-off between the quality of the pooling and the depth of the features that must be accounted for.

6.2. Study of deformable attention

1) Comparison w.r.t. other baselines: Here we empirically evaluate the benefits of our proposed RDSA. Concretely, in Tab. 5 we empirically compare the performance of the standard CA [35] and Def. CA [45], as well as different decoder-query initialization strategies [43] for the latter. Notice the prominent performance degradation de-

Att. strat.	MR					HD	
	R1@0.5	R1@0.7	mAP@0.5	mAP@0.75	mAP	mAP	HIT@1
Stand. CA [35]	70.77	51.74	66.78	45.19	42.72	43.16	69.87
Def. CA [45]	72.58	58.19	71.82	55.80	54.27	43.26	70.58
Def. CA +PureInit [43]	72.13	57.87	70.68	54.85	52.92	43.19	70.32
Def. CA +MixedInit [43]	72.26	58.71	71.94	56.43	54.94	43.10	69.86
Ours	**73.68**	**60.90**	**73.53**	**57.42**	**55.60**	**44.0**	**72.00**

Table 5. Comparison across different attention strategies as well as various decoder query initializations.

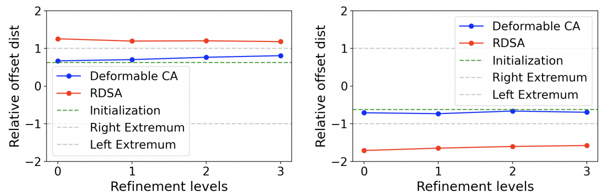


Figure 4. Average of the weighted offsets across M decoder queries and N batch elements for the $K = 4$ refinement levels. Here head 0 (left) is initialized near the left boundary, and head 1 (right) near the right boundary.

rived from the use of the CA, or how the query initialization techniques –far from improving the performance of the Def. CA– in some cases are even harmful. In contrast, our method consistently outperforms all these tested baselines, improving by 2.71% R1@0.7 or 1.62% mAP w.r.t. the original Def. CA. We refer to Sec. K for the ablation of the different sampling strategies of RDSA.

2) Where are the offsets pointing to? In Fig. 4 we depict the refinement of the weighted offset of a given query q, defined as $d_q = \sum_{p=1}^{P} A_{q,p}\Delta_{q,p}$, across the K different refinement steps. Note that as depicted in Eq. 9, these offsets are relative to the estimated moment widths, with -1 and +1 indicating the estimated left- and right-most boundaries. The Def. CA lacks context awareness, causing its offset predictions to stay near the offset initializations. In contrast, RDSA leverages an enhanced contextualization, thus reducing its bias towards the offset initialization. Notably, RDSA learns to *look* beyond the current action boundaries (offsets < -1 and $> +1$) capturing near-boundary information, crucial for refining moment boundaries. This contrasts with [45] which remains constrained within the predicted moment boundaries. We consider this the reason why RDSA is especially useful to localize long actions (see Sec. K).

6.3. Conditioning signal for the sparse stream $\mathcal{S}$

To effectively integrate RDSA, it is imperative to obtain an expressive video-based conditioning signal –i.e., the red arrow in Fig. 2a. As shown in Tab. 6, using the raw video representation $\mathbf{V}^\ell$ significantly degrades performance, highlighting the need for a richer signal incorporating textual and temporal information. While conditioning on the outputs of the CA or SA modules of the dense stream $\mathcal{D}$ improves results, these lack non-linearities. Instead, using the non-linear PFFN output provides greater flexibility, leading to superior expressiveness. Notably, this choice also enhances HD performance, underscoring the benefits of task

Type of features	MR					HD	
	R1@0.5	R1@0.7	mAP@0.5	mAP@0.75	mAP	mAP	HIT@1
Raw. rep	72.58	56.37	70.12	50.99	50.74	44.13	71.37
Post CA	72.26	58.39	72.20	55.92	54.83	43.64	70.65
Post CA+SA	72.13	58.32	72.00	56.57	54.94	43.78	70.06
Post CA+SA+FFN	**73.68**	**60.90**	**73.53**	**57.42**	**55.60**	**44.00**	**72.00**

Table 6. Evaluation of various conditioning signals.

CA	SA	RDSA	FFN	MR					HD	
				R1@0.5	R1@0.7	mAP@0.5	mAP@0.75	mAP	mAP	HIT@1
	✓	✓	✓	71.42	57.74	71.73	56.18	54.47	43.53	71.35
✓		✓	✓	71.87	58.71	72.13	55.99	54.32	43.28	69.55
✓	✓		✓	59.68	30.71	59.02	24.6	29.09	43.58	70.39
✓	✓	✓		72.84	59.29	73.13	56.91	55.72	43.85	72.58
✓	✓	✓	✓	**73.68**	**60.90**	**73.52**	**57.42**	**55.60**	**44.0**	**72.00**

Table 7. Importance of the modules of the sparse stream $\mathcal{S}$.

interaction between MR and HD.

6.4. Effect of each module of the sparse stream $\mathcal{S}$

In this ablation, we empirically evaluate the contribution of each of the components of $\mathcal{S}$. Concretely, we evaluate the influence of the CA, the SA, the RDSA, and the final PFFN. Observe in Tab. 7 that the initial textual injection CA seems to be particularly relevant for the R1 metric, while the reasoning SA module is important for the mAP. Moreover, as expected, the RDSA module is critical. This is to be expected, as without it, the queries become oblivious to the input video, leaving no other option but that of randomly guessing. Again, the additional expressivity that the final PFFN brings via its non-linearities proves to be beneficial, despite affecting less than the other modules.

7. Conclusions

In this work, we introduced the *Sparse-Dense Side-Tuner*, the first proposal-free side-tuning method for VTG. Our dual-stream architecture jointly optimizes dense embeddings and *recurrent decoder queries*, carefully adapting to the sparse and dense nature of MR and HD. We also identify an inherent context limitation of the Deformable CA –a key component of proposal-free architectures like ours– which motivates our proposed alternative, the *Reference-based Deformable Self-Attention*. Finally, we provide the first effective integration of InternVideo2 to a side-tuning framework, leading to a substantial performance boost. We evaluate our method on QVHighlights, TACoS, and Charades-STA, surpassing previous ST works and attaining either SOTA or near-SOTA, all while maintaining a minimal parameter count and a low memory overhead. We leave as future work studying the often overlooked problem of the cross-domain generalization capabilities of these methods.

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